SOFTWARE REQUIREMENTS SPECIFICATION

for

BLOODLINE

Predictive Blood Shortage and Smart Donor Matching Platform

Version 1.0

Prepared by :
1. Avraj Singh Sekhon (1024030287)
2. Krish Kumar (1024030296)
3. Kyna Sood (1024030303)
4. Manya Sabherwal (1024030288)

Submitted to : Ms. Anushka
Course UCS503P
Thapar Institute of Engineering and Technology

October 13, 2026

Contents

1 Introduction **3**
- 1.1 Purpose . . . 3
- 1.2 Intended Audience and Reading Suggestions . . . 3
- 1.3 Project Scope . . . 3

2 Overall Description **5**
- 2.1 Product Perspective . . . 5
- 2.2 User Classes and Characteristics . . . 5
- 2.3 Product Functions . . . 6
- 2.4 Operating Environment . . . 7
- 2.5 Design . . . 7

3 System Features **9**
- 3.1 Description and Priority . . . 9
- 3.2 Functional Requirements . . . 9

4 Other Nonfunctional Requirements **13**
- 4.1 Performance Requirements . . . 13
- 4.2 Security Requirements . . . 13
- 4.3 Software Quality Attributes . . . 13
- 4.4 Business Rules . . . 14

5 Other Requirements **15**

1 Introduction

1.1 Purpose

Blood banks and hospitals often find out that a blood group is running low only when the shortage has already started. Stock is tracked, but future demand is not forecast; blood-bank stock, hospital needs and donors are kept in different places; and finding suitable donors by blood group, donation history, distance, availability and consent is slow manual work. So, "BloodLine" is the solution.
"BloodLine" is a web-based platform which combines inventory tracking, short-term demand forecasting, shortage-risk detection and rule-based donor matching in one role-based system. The main concept of "BloodLine" is to change blood availability management from a reactive process into a proactive one. This SRS describes the proposed system of the project proposal for UCS503P.

1.2 Intended Audience and Reading Suggestions

This SRS is for developers, project managers, users, testers and course evaluators. Further the discussion will provide all the internal, external, functional and also non-functional information about "BloodLine". Requirements are marked **MVP** (first iteration) or **Inc. 2** (subsequent deliverables).

1.3 Project Scope

"BloodLine" creates a space for Donors, Blood-Bank Managers, Inventory Staff, Medical Staff, Hospital Coordinators and System Administrators for managing blood availability.
Inventory staff record the blood stock by blood group and every receipt, issue and expiry. BloodLine forecasts near-term demand from historical data and compares it with the stock. Every blood group gets a LOW, MODERATE or HIGH shortage-risk flag with a plain-English reason. The manager reviews the risk and asks BloodLine for matching donors: compatible blood group, past the donation cooldown, available, nearby and with notification consent. Donors are only suggested. Medical staff do the final screening before a donation is recorded.
Hospital coordinators send blood requests. The manager approves or rejects them, and an approved request issues the units from inventory. The administrator manages accounts and roles.
BloodLine does not make final medical eligibility decisions and does not move blood between banks. Forecasts are risk signals, not guarantees.

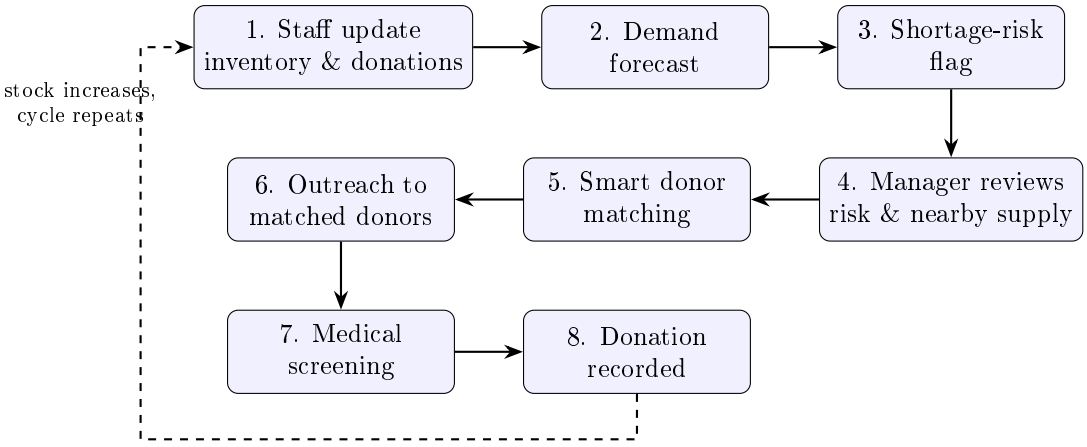


Figure 1.1: Entire work-flow

Figure 1.1 (Entire work-flow) is the overview of the project. Connection of all the entities are dependable to each other. This gives the simple idea about the functional activities of the project.
Inventory cycle, in Figure 1.1: "Staff" update "Inventory"; BloodLine "forecasts" and raises a "risk flag"; the "Manager" reviews it.
Donor cycle, in Figure 1.1: the "Manager" asks for "matches"; "Donors" are contacted; "Medical staff" screen them; the "Donation" is recorded and stock rises.

2 Overall Description

2.1 Product Perspective

"BloodLine" is the replacement of the manual and scattered blood availability process. Stock registers, donor lists and request slips are kept in one system. Main goal of this project is to minimize the time needed to find and respond to a shortage. It is a new web application with a 3-tier architecture (Figure 2.1).

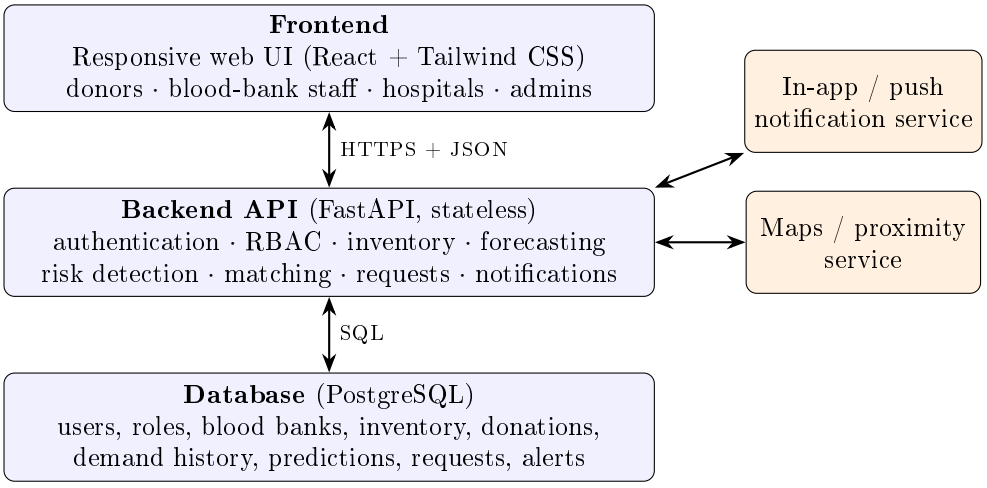


Figure 2.1: Three-tier system architecture

2.2 User Classes and Characteristics

"BloodLine" has basically 6 types of users. Each user has one account with one role, and the role decides what he/she can see and do.

- Donors
- Hospital Coordinators
- Blood-Bank Staff
  - Blood-Bank Manager
  - Inventory Staff
  - Medical Staff
- System Administrators

Donor keeps own profile, availability and consent. Manager reviews risk, finds donors and approves requests. Inventory staff keep the stock correct. Medical staff screen donors. Hospital coordinator sends requests of own hospital. Administrator manages accounts.

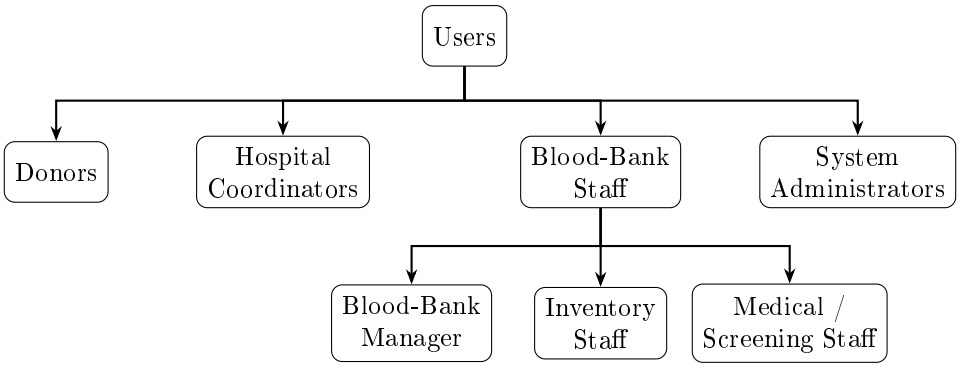


Figure 2.2: Type of users

2.3 Product Functions

"BloodLine" stores the inventory, donors, demand history, forecasts and requests of blood banks. Before using the main functions, users have to be registered.
All users have - id, email, hashed_password, full_name, role, is_active and created_at.
Donors have some extra information - user_id (foreign key), blood_group, last_donation_date, latitude, longitude, is_available and notification_consent.
Blood banks have - name, address, latitude and longitude. Inventory has blood_group, component and quantity_units; every stock change is an inventory transaction (receipt, issue, expiry or discard).
Demand history has units_demanded per blood group per day, with an is_synthetic flag. Predictions and alerts store the forecast, risk_level, shortfall and explanation.
Hospital requests have blood_group, quantity_units, urgency (routine, urgent, emergency) and status. Screening records have donor, staff member, outcome (eligible, deferred, rejected) and notes.

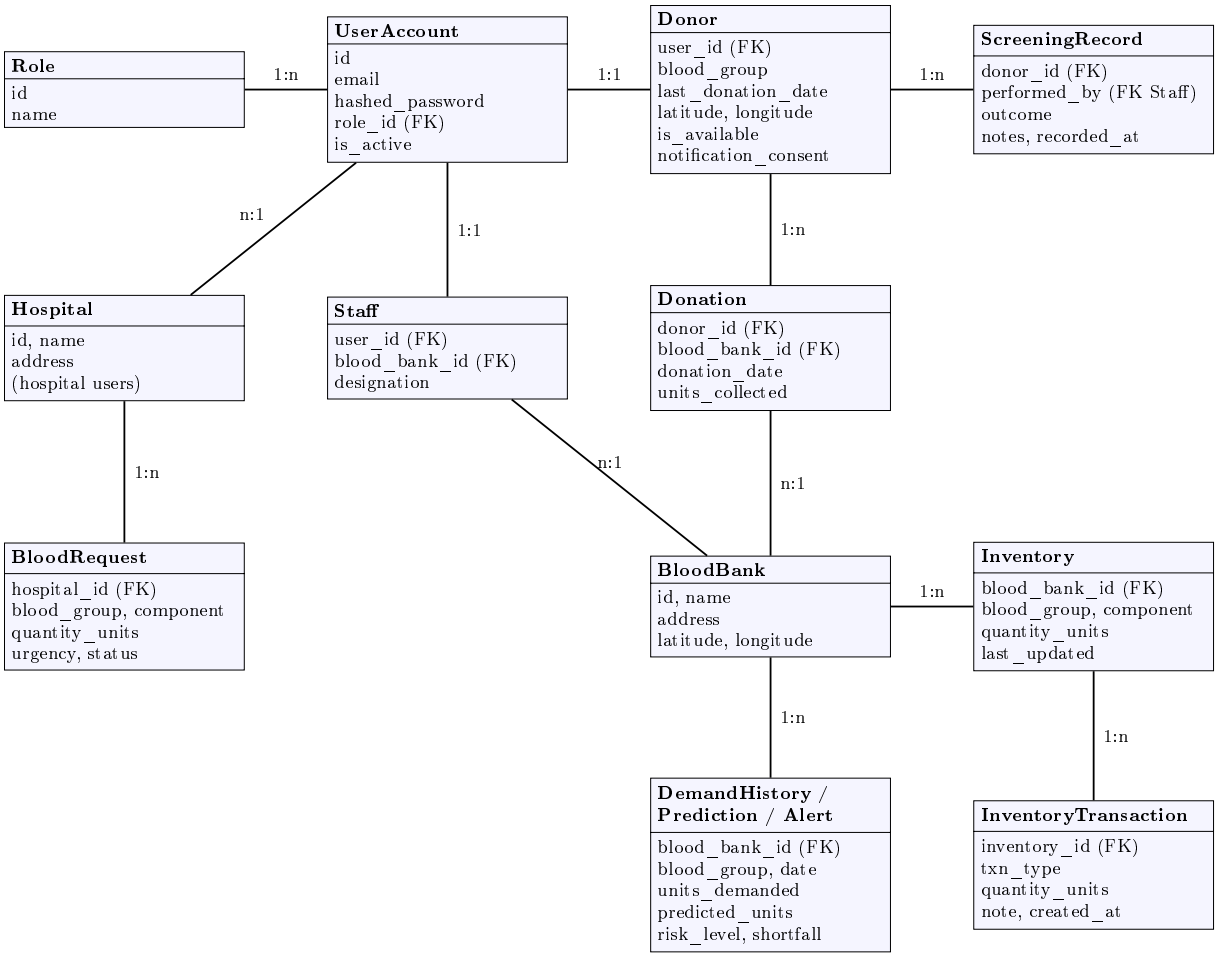


Figure 2.3: Data model (main entities and relations)

Alert is the main feature of all. It combines the stock of a blood group with its forecast and shows the manager the level, the shortfall and the reason.

2.4 Operating Environment

The website will be operated in any Operating Environment - Mac, Windows, Linux, Android, iOS etc. - through a modern web browser. The backend runs on common web hosting with a managed database. The interface is responsive, so low-to-mid range devices can use it. Where real consumption data is not available, clearly labelled simulated data is used.

2.5 Design

Donor activities have 3 steps - Registration and Profile, Availability and Consent, Alerts and Donation History. Activities of the other roles are shown in Figure 2.4.

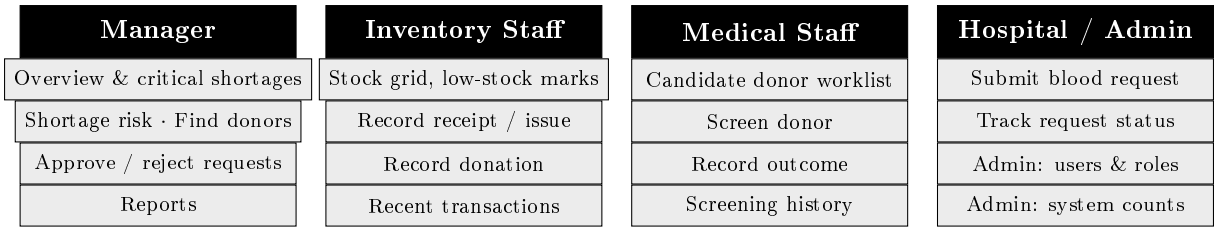


Figure 2.4: Role-wise activities of staff, hospital and admin users

Manager can review the risk, find donors, and approve or reject requests. Inventory staff record stock and donations. Medical staff only screen donors. Hospital coordinator only creates and tracks requests. Administrator only manages accounts.

3 System Features

"BloodLine" is a predictive blood-availability web software. So the main art of this product is to keep stock correct, forecast demand, detect shortage risk early and match donors.

3.1 Description and Priority

"BloodLine" has features that are main and also some are sub. But all the features are necessary for this software.
The features with priority up to down -

1. Inventory Tracking : The base of all. Operated by inventory staff. MVP
2. Demand Forecasting : Estimates near-term demand of every blood group. MVP
3. Shortage-Risk Detection : Compares forecast with stock and flags risk. This is the goal feature. MVP
4. Smart Donor Matching : Suggests compatible, available and consenting donors. MVP
5. Role-Based Access : Separate pages and permissions for the six roles. MVP
6. Hospital Requests : Request, approval and automatic issue of stock. Inc. 2
7. Screening Workflow : Medical staff clear donors and record outcomes. Inc. 2
8. Notifications : Alerts only to matched donors. Inc. 2
9. Reports and Regional View : Summaries and heatmap of surplus and shortage. Inc. 2

3.2 Functional Requirements

Each requirement has an ID (FR-xxx-n), a "shall" statement and the increment (**M** = MVP, **2** = Inc. 2).

Authentication and Access (AUTH)

ID	Requirement	Inc.
FR-AUTH-1	The system shall let a registered user log in with email and password; passwords are stored only as salted hashes.	M
FR-AUTH-2	On login the system shall issue a signed token with the user's role, which expires after a fixed time (default 8 hours).	M
FR-AUTH-3	The system shall deny any request without a valid token or from a role that is not authorised, and shall block deactivated accounts.	M

ID	Requirement	Inc.
FR-AUTH-4	A donor shall be able to self-register with blood group, approximate location and consent.	2

Inventory (INV)

ID	Requirement	Inc.
FR-INV-1	The system shall keep current stock (units) per blood bank, blood group and component.	M
FR-INV-2	Inventory staff shall record receipt, issue, expiry or discard with quantity and note; every transaction is timestamped and logged.	M
FR-INV-3	The system shall reject zero or negative quantities and any issue larger than the units available.	M
FR-INV-4	The dashboard shall mark blood groups below a low-stock level (default 15 units).	M
FR-INV-5	A recorded donation shall update the donor's last donation date and increase the stock.	M
FR-INV-6	The system shall track single units with collection and expiry date and flag units close to expiry.	2

Demand Forecasting (FCST)

ID	Requirement	Inc.
FR-FCST-1	The system shall store daily demand per blood group; simulated data shall carry an is_synthetic flag and be labelled "Synthetic / Simulated" on every screen.	M
FR-FCST-2	The system shall forecast near-term demand (default 7 days) of every blood group using a time-series model (Prophet or SARIMA in the final system).	M
FR-FCST-3	Every forecast shall be compared with a naive moving-average baseline and shall report the model used, the predicted units and the error.	M
FR-FCST-4	Data shall be split chronologically into training and test periods, and MAE and RMSE computed on the test period.	M
FR-FCST-5	The model shall be replaceable without changing the API, and the data layer shall accept authorized real data.	2

Shortage Risk (RISK)

Shortfall = forecast demand − current stock.

ID	Requirement	Inc.
FR-RISK-1	The system shall classify risk as **LOW** if shortfall ≤ 0; **MODERATE** if $0 <$ shortfall $\leq 25\%$ of stock; **HIGH** if shortfall $> 25\%$ of stock, or if stock is zero and demand is expected. Thresholds are configurable.	M
FR-RISK-2	Every result shall carry a plain-English explanation (stock, forecast, shortfall, reason); the rule layer is transparent, not a black box.	M
FR-RISK-3	Alerts shall be stored with level, shortfall and timestamp, and the manager's overview shall list critical shortages with a "Find donors" action.	M

Example: 20 units in stock and 25 forecast gives shortfall 5, which is 25% of stock, so MODERATE.

Donor Matching (MATCH)

ID	Requirement	Inc.
FR-MATCH-1	For a target blood group the system shall include only compatible donor groups (table below).	M
FR-MATCH-2	The system shall exclude donors who are unavailable, have not given consent, or donated within the cooldown (default 90 days).	M
FR-MATCH-3	The system shall filter donors by distance within a configurable radius.	2
FR-MATCH-4	Candidates shall be ranked by a score (exact group match, longer time since last donation, nearness), each with reasons.	M
FR-MATCH-5	Results shall be shown as candidates to contact; the system shall never declare a donor medically eligible.	M

Patient	Compatible donors	Patient	Compatible donors
O−	O−	O+	O−, O+
A−	O−, A−	A+	O−, O+, A−, A+
B−	O−, B−	B+	O−, O+, B−, B+
AB−	O−, A−, B−, AB−	AB+	all eight groups

Requests, Screening, Notifications, Reports (Inc. 2)

ID	Requirement	Inc.
FR-REQ-1	A hospital coordinator shall submit a request (group, quantity, urgency, notes) and see only the own hospital's requests.	2

ID	Requirement	Inc.
FR-REQ-2	On approval the system shall issue the units from stock and set the status *fulfilled*; if stock is not enough the request stays *pending* with a note. The manager can also reject.	2
FR-SCR-1	Medical staff shall see a worklist of candidate donors and record the outcome (eligible, deferred, rejected) with notes. The outcome is always entered by a person, never automatically.	2
FR-NOTE-1	The system shall send in-app/push alerts only to donors who satisfy the matching rules, store them, and stop when consent is withdrawn.	2
FR-RPT-1	The manager shall view reports of units received, issued and expired, risk summary and demand versus forecast; reports shall be exportable.	2
FR-REG-1	The system shall show a regional heatmap of surplus and shortage across blood banks to help coordination.	2

Donor Portal and Administration (MVP)

ID	Requirement	Inc.
FR-DON-1	A donor shall view own profile, eligibility status and donation history, and switch availability and consent.	M
FR-ADM-1	The administrator shall view system counts and users, create and deactivate accounts, and assign roles.	M
FR-ADM-2	The system shall write basic logs with timestamps for logins, stock changes and admin actions.	M

Technology Stack

The "BloodLine" website is being built on React, Tailwind CSS, FastAPI (Python), PostgreSQL and Pandas with statsmodels/Prophet.
Front-End - React.js, Tailwind CSS, JavaScript.
Back-End - FastAPI (Python), REST API.
Database - PostgreSQL.
Forecasting - Pandas, statsmodels, Prophet. Maps - Google Maps API or equivalent.

4 Other Nonfunctional Requirements

4.1 Performance Requirements

"BloodLine" will be used in daily work of blood banks, so common actions must be quick. Targets for up to 50 concurrent users: inventory update $\leq$ 1 s, dashboard loading $\leq$ 3 s, donor search $\leq$ 2 s. For growth, the backend is stateless, frontend and backend are decoupled, and large lists use pagination and database indexes.

4.2 Security Requirements

No one without registered users can enter the website. One particular user of a role only can perform his/her particular actions.

- Passwords are stored as salted hashes; sessions use signed, expiring tokens; all traffic uses HTTPS.
- Role checks are done on the server for every protected request.
- A donor sees only own data; a hospital coordinator sees only own hospital's requests.
- Personal data is minimised, and alerts go only to consenting donors who satisfy the matching rules.

4.3 Software Quality Attributes

In the development phase also testing and conferences of users is continued, so that the quality of the software is maintained and all the requirements are fulfilled. Database, logical and also UI test is required. Success is measured by:

Metric	**How it is measured**
Forecasting error (primary)	MAE and RMSE of predicted versus actual demand in the held-out test period
Shortage-risk performance	Known shortage cases in test data are correctly flagged
Donor-matching correctness	Seeded donors check that group, cooldown, proximity, availability and consent filters work
Workflow response time	Time of inventory update, dashboard load and donor search
Notification targeting	Alerts reach only donors who satisfy the matching rules

Other attributes: stock never becomes negative; every risk level and match has a readable reason; forecasting, risk and matching are separate modules, so a model can be swapped; the build runs automated unit tests and linting on every change.

4.4 Business Rules

"BloodLine" is for finding and preventing blood shortages early. Basically save working time and pressure.

- Final donor clearance is given only by qualified medical staff.
- Actual movement of blood between banks remains with authorized staff.
- A donor is contacted only if compatible, available, consenting and past cooldown.
- A request is fulfilled only if enough stock exists.
- Simulated data is never presented as real data; forecasts always get human review.

5 Other Requirements

"BloodLine" needs maintenance as it is a long process software. It will need re-factoring and further the requirements can be changed as the field is changing frequently. CI/CD will build and test every change automatically and deploy to staging, so features are released in weekly iterations. Main risks are handled as - unavailable real data: use labelled simulated data and keep the data layer ready for real data; forecast uncertainty: show risk signals with human review; medical and privacy limits: staff do final clearance and access is by role.